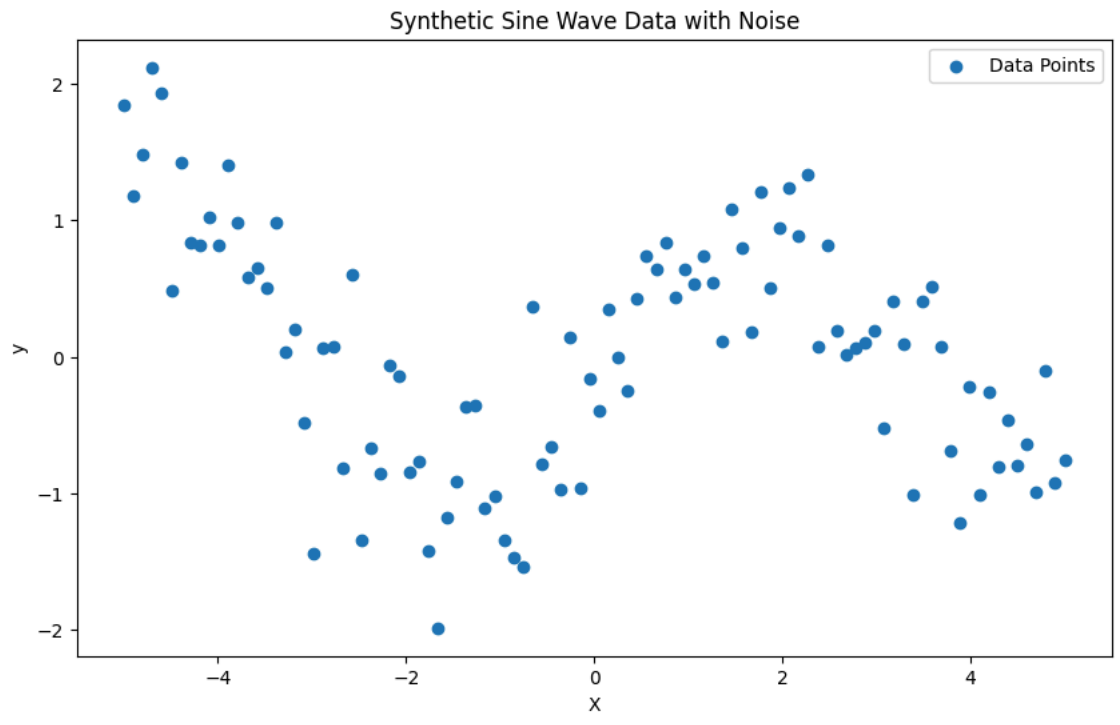
Module 08 Lab - The Bias-Variance Tradeoff**Objective:** To understand and visualize the concepts of **overfitting**, **underfitting**, and the **bias-variance tradeoff**, which are central to building models that generalize well to new data.**In this lab, you will train models of varying complexity and plot their performance to see these concepts in action.**

Part 1: Understanding the Concepts* **Underfitting (High Bias):** The model is **too simple** and fails to capture the underlying patterns in the data. It performs poorly on both the training data and the test data. It has high bias because it makes strong, incorrect assumptions about the data.* **Overfitting (High Variance):** The model is **too complex** and learns the training data too well, including the noise and random fluctuations. It performs exceptionally well on the training data but poorly on the test data because it has memorized the training set instead of learning the general pattern. It has high variance because its performance changes drastically with different training data.* **The Goal:** Find a model that is "just right"—complex enough to capture the true pattern but not so complex that it memorizes the noise. This is the **bias-variance tradeoff**.

Part 2: SetupTo visualize this, we will create a synthetic (fake) dataset with a known pattern and some noise.

```
import matplotlib.pyplot as plt
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import learning_curve
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import PolynomialFeatures

# 1. Generate synthetic data
np.random.seed(0)
X = np.linspace(-5, 5, 100)
y = np.sin(X) + np.random.normal(0, 0.5, 100)

# Reshape X for scikit-learn estimators
X = X[:, np.newaxis]

# 2. Plot the data to visualize the underlying pattern
plt.figure(figsize=(10, 6))
plt.scatter(X, y, label="Data Points")
plt.title("Synthetic Sine Wave Data with Noise")
plt.xlabel("X")
plt.ylabel("y")
plt.legend()
plt.show()
```



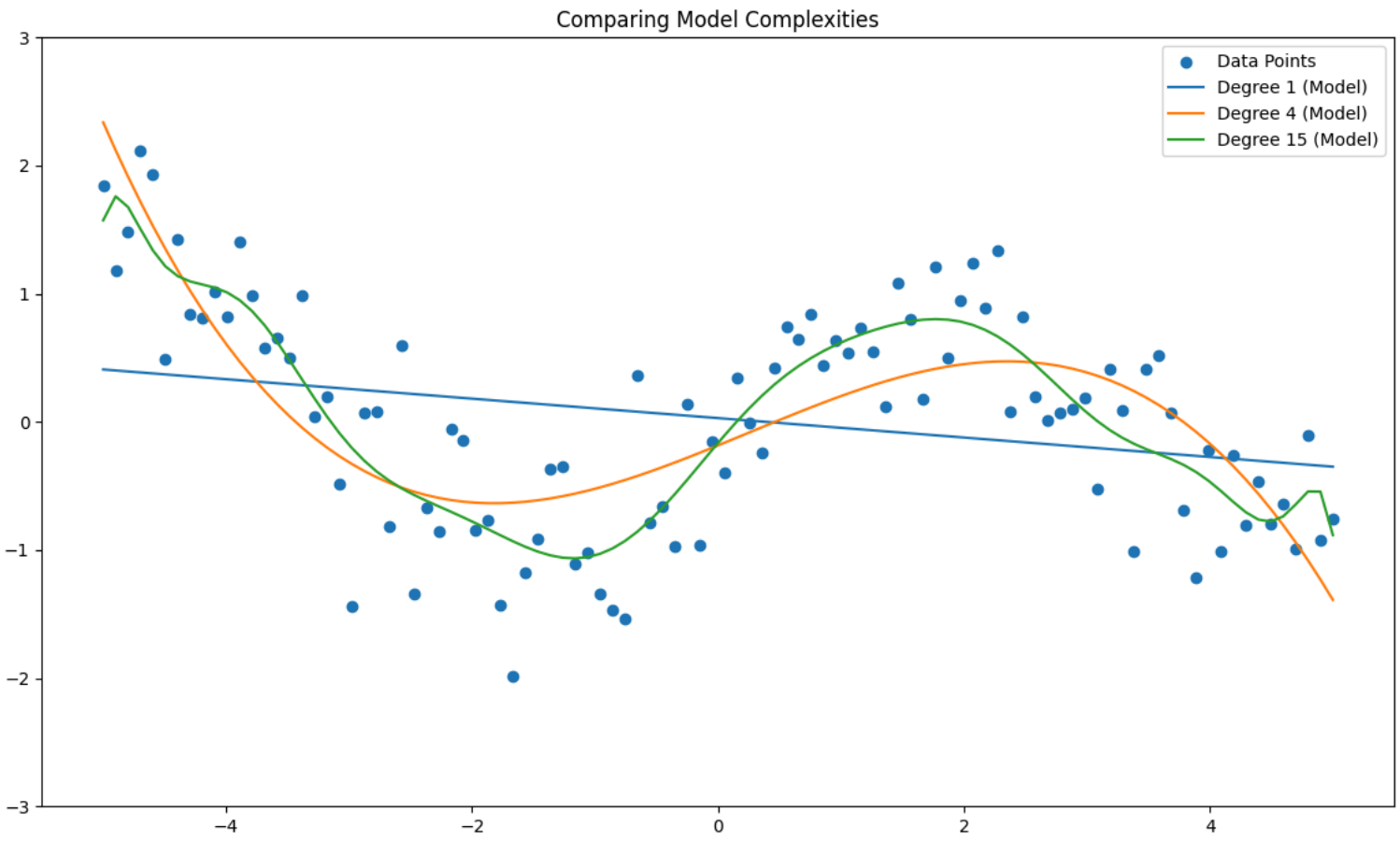
Part 3: Modeling with Different ComplexitiesWe will use **Polynomial Regression** to control model complexity. A polynomial of degree 1 is a simple straight line (underfitting). A polynomial of degree 15 is a very complex, wiggly line (overfitting).**Your Task:** Train and visualize three models with different degrees (1, 4, and 15) to see underfitting, a good fit, and overfitting.

Task 1: Train and Plot Models**

Your Task:** For each degree (1, 4, 15), create a polynomial regression model, train it, and plot its predictions against the original data.

```
# --- ENTER YOUR CODE HERE ---
plt.figure(figsize=(14, 8))
plt.scatter(X, y, label="Data Points")

degrees = [1, 4, 15]

for degree in degrees:
    # 1. Create a pipeline that adds polynomial features and then fits a linear regression model    #
    model = make_pipeline(PolynomialFeatures(degree), LinearRegression())

    # 2. Train the model    #
```

```
    model.fit(X, y)

    # 3. Make predictions    #
    y_pred = model.predict(X)

    # 4. Plot the model's predictions    #
    plt.plot(X, y_pred, label=f"Degree {degree} (Model)")

plt.legend()
plt.title("Comparing Model Complexities")
plt.ylim(-3, 3)
plt.show()
```

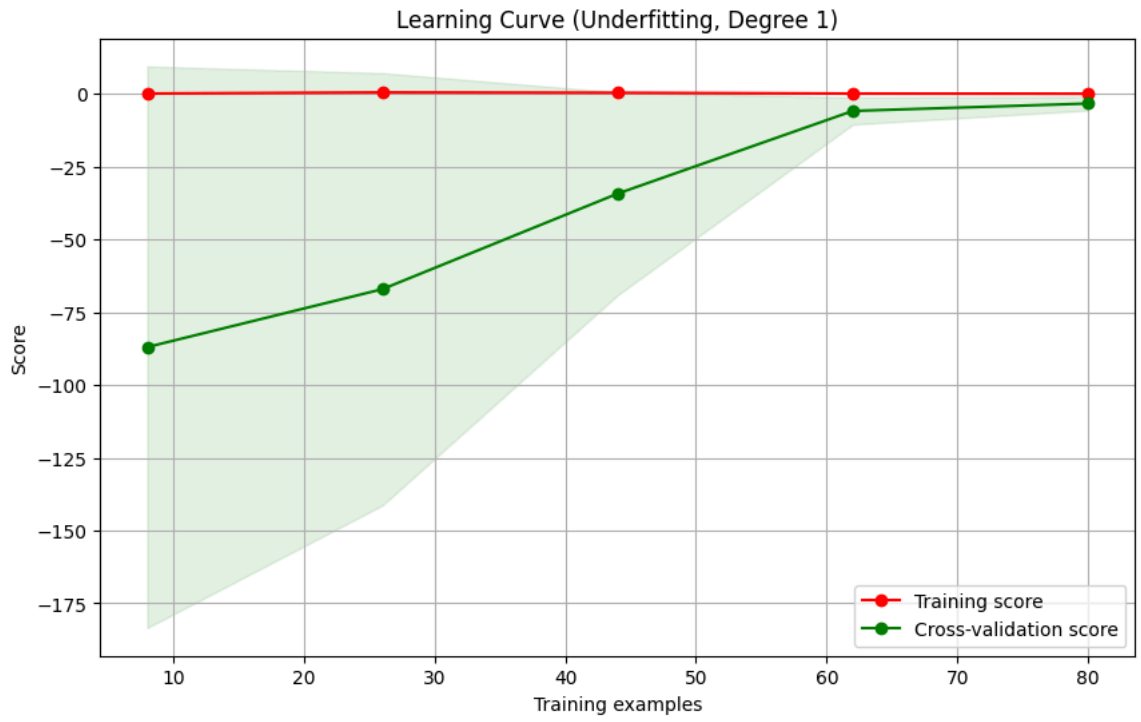


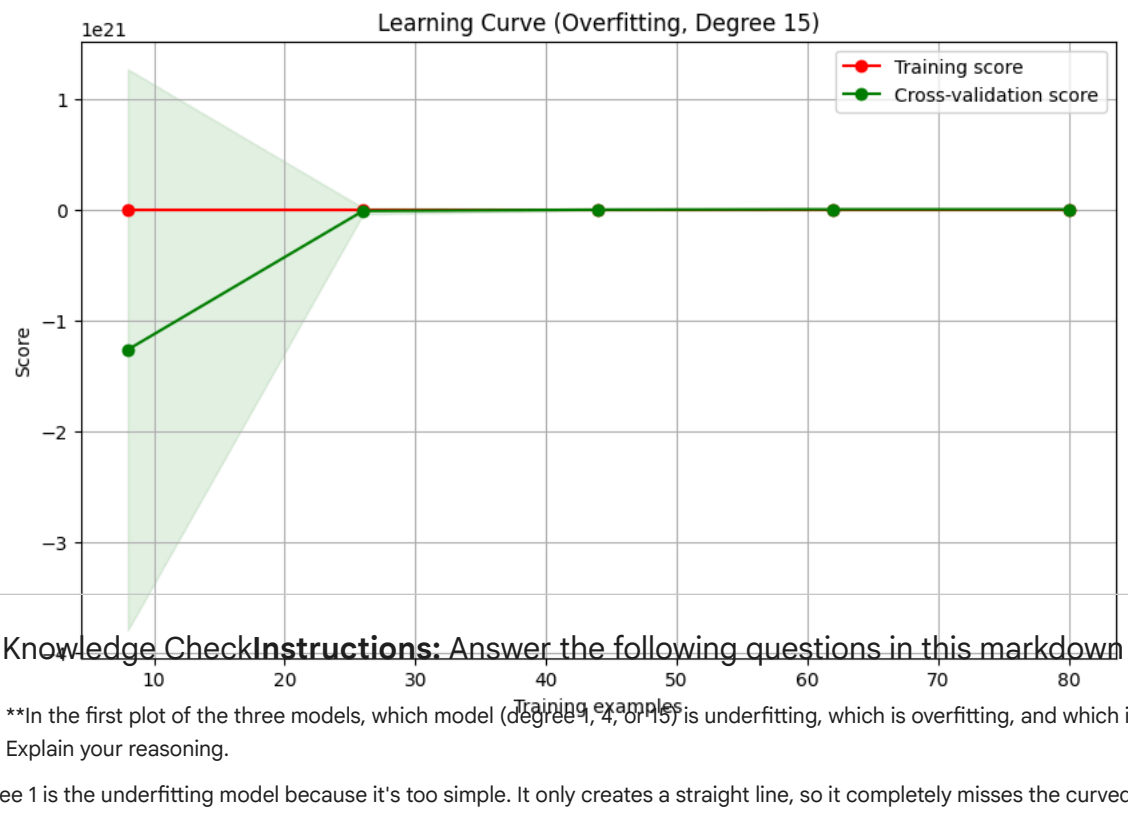
Part 4: Learning Curves**Concept:** A **learning curve** is a powerful tool to diagnose bias and variance. It plots the model's performance (e.g., accuracy or error) on both the **training set** and the **validation set** as a function of the number of training samples.* **High Bias (Underfitting):** Both the training score and validation score will be low and will plateau quickly. The model is too simple to learn from more data.* **High Variance (Overfitting):** There will be a large gap between the high training score and the low validation score. The model memorized the training data but can't generalize.* **Just Right:** The training and validation scores will converge to a high value.

Task 2: Plot Learning Curves**Your Task:** Use the `learning_curve` function from scikit-learn to plot the learning curves for the underfit (degree 1) and overfit (degree 15) models.

```
def plot_learning_curve(
    estimator,
    title,
    X,
    y,
    ylim=None,
    cv=None,
    n_jobs=None,
    train_sizes=np.linspace(0.1, 1.0, 5),
):
    plt.figure(figsize=(10, 6))
    plt.title(title)
    if ylim is not None:
        plt.ylim(*ylim)
    plt.xlabel("Training examples")
    plt.ylabel("Score")
    train_sizes, train_scores, test_scores = learning_curve(
        estimator, X, y, cv=cv, n_jobs=n_jobs, train_sizes=train_sizes
    )
    train_scores_mean = np.mean(train_scores, axis=1)
    train_scores_std = np.std(train_scores, axis=1)
    test_scores_mean = np.mean(test_scores, axis=1)
    test_scores_std = np.std(test_scores, axis=1)
    plt.grid()
    plt.fill_between(
        train_sizes,
        train_scores_mean - train_scores_std,
        train_scores_mean + train_scores_std,
        alpha=0.1,
        color="r",
    )
    plt.fill_between(
        train_sizes,
        test_scores_mean - test_scores_std,
        test_scores_mean + test_scores_std,
        alpha=0.1,
        color="g",
    )
    plt.plot(
        train_sizes,
        train_scores_mean,
        "o-",
        color="r",
        label="Training score",
    )
    plt.plot(
        train_sizes,
        test_scores_mean,
        "o-",
        color="g",
```

```
        label="Cross-validation score",
    )
    plt.legend(loc="best")
    return plt


# --- ENTER YOUR CODE HERE ---

# 1. Create the underfit and overfit models
underfit_model = make_pipeline(PolynomialFeatures(1), LinearRegression())
overfit_model = make_pipeline(PolynomialFeatures(15), LinearRegression())

# 2. Plot the learning curve for the underfit model
plot_learning_curve(
    underfit_model, "Learning Curve (Underfitting, Degree 1)", X, y, cv=5
)
plt.show()

# 3. Plot the learning curve for the overfit model
plot_learning_curve(
    overfit_model, "Learning Curve (Overfitting, Degree 15)", X, y, cv=5
)
plt.show()
```





📝 Knowledge Check**Instructions:** Answer the following questions in this markdown cell.

1. **In the first plot of the three models, which model (degree 1, 4, or 15) is underfitting, which is overfitting, and which is a good fit? Explain your reasoning.

Degree 1 is the underfitting model because it's too simple. It only creates a straight line, so it completely misses the curved patterine in the data. Where as Degree 4 is the good fit because it follow the overall shape of the data without trying to fit every random point. I captrues the patter while still generalining well. Then Degree 15 is the overfitting model because it't too complex. Instead of learning the overal trend, its start fintting th random noise in the training data.

2. **Looking at the learning curve for the underfitting model, what do you observe about the training and cross-validation scores? What does this tell you?

The training and cross-vallidation scores stay colse together, but they're both low. That indicates that the model isn't learing the pattern very well. Since the model is too simple, adding more data would not improve it. It would require more complex model instead.

3. **Looking at the learning curve for the overfitting model, what do you observe about the gap between the training and cross-validation scores?** What does this tell you?**[ENTER YOUR ANSWERS HERE]**

The training score is very high, but the cross-validation score is much lower, so there;s a big gap between them. This indicated the model learned the data too well, inculding the random noise. Because of that, it doesn't perform well on new data, which is a sign of high variance.